

Smart Traffic Signal Control Using Local AI for Emergency Vehicle Priority

KHM Niveditha

Department of Computer Science and Engineering - Data Science
Ballari Institute of Technology and Management
Ballari, Karnataka, India

Abstract—Traffic congestion at road intersections can delay emergency vehicles such as ambulances and fire engines. Conventional fixed-time traffic signals do not continuously adapt to changing traffic conditions, while intelligent traffic signal control methods can use traffic information to modify signal operation. This paper presents a conceptual local-AI-assisted traffic signal architecture in which roadside cameras provide traffic information to a local computing unit. The local unit estimates traffic conditions and detects an emergency vehicle when suitable visual information is available. Based on the detected condition, the traffic signal controller can adjust the signal phase to provide priority to the emergency vehicle while considering the effect on other traffic. The paper reviews existing research on adaptive traffic signal control, machine vision, and emergency vehicle preemption, and proposes a system architecture and evaluation framework for future implementation.

Index Terms—Traffic Signal Control, Local AI, Computer Vision, Emergency Vehicle Preemption, Intelligent Transportation Systems

I. INTRODUCTION

Traffic congestion at urban intersections can increase vehicle waiting time and create difficulties for emergency vehicles. Traffic signal control is therefore an important component of intelligent transportation systems.

Traditional traffic signals commonly operate according to predefined timing plans. Such systems are simple to operate, but their fixed operation does not directly respond to changes in traffic demand. Research on intelligent traffic signal control has investigated machine learning and deep reinforcement learning approaches for adapting signal decisions to traffic conditions [1].

Machine-vision-based traffic monitoring provides another approach. Vehicle information obtained from cameras can be used to estimate traffic conditions and support adaptive signal timing [2].

Emergency vehicle preemption introduces an additional requirement. When an ambulance, fire engine, or other emergency vehicle approaches an intersection, the signal controller may need to modify the normal signal sequence. Previous research has investigated priority algorithms and emergency-vehicle-centered signal control to reduce emergency vehicle delay while limiting disruption to other traffic [3], [4].

This paper proposes a conceptual architecture that combines local traffic observation with emergency vehicle priority. The objective is not to claim experimental performance, but to

describe a technically testable system and an evaluation procedure.

II. RELATED WORK

A. AI-Based Traffic Signal Control

Rasheed et al. reviewed the application of deep reinforcement learning to traffic signal control and discussed different architectures, simulation platforms, design considerations, and open research problems [1]. Their review shows that learning-based traffic signal control can be investigated as an alternative to conventional signal-control strategies.

B. Machine Vision for Adaptive Signals

Gorodokin et al. investigated adaptive traffic-light control using machine vision. Their work considered vehicle-related information, including traffic flow and queue characteristics, when adjusting traffic signal operation [2]. This demonstrates how camera-based traffic information can support adaptive signal control.

C. Emergency Vehicle Priority

Asaduzzaman and Vidyasankar proposed a priority algorithm for emergency vehicle traffic signal control. Their study used the VISSIM microscopic traffic simulator to evaluate emergency vehicle priority and reported a reduction in overall traffic delay compared with a conventional emergency vehicle preemption approach [3].

Cao, Shuai, and Li investigated emergency vehicle-centered traffic signal control with the objective of reducing emergency vehicle delay while limiting the effect on conflicting traffic [4].

These studies provide the foundation for combining traffic-state estimation with emergency vehicle priority in the proposed architecture.

III. PROBLEM STATEMENT

Fixed-time traffic signals do not directly respond to changes in traffic demand or emergency situations. Adaptive traffic signal systems can respond to traffic information, but an emergency vehicle requires a specific priority mechanism.

Therefore, the problem considered in this paper is the design of a traffic signal architecture that can:

- obtain traffic information from roadside cameras;
- process relevant information locally;

- identify an emergency vehicle when sufficient visual evidence is available;
- determine an appropriate signal-control action; and
- provide emergency priority while considering surrounding traffic.

IV. OBJECTIVES

The objectives of the proposed system are:

- To monitor traffic conditions using roadside cameras.
- To perform traffic analysis close to the intersection.
- To identify emergency vehicle situations from available visual information.
- To generate an appropriate traffic signal priority request.
- To evaluate emergency vehicle delay and the effect on other traffic.

V. PROPOSED SYSTEM

The proposed system uses roadside cameras and a local computing unit to obtain and process traffic information. The local unit acts as an intermediate processing layer between the cameras and the traffic signal controller.

The system consists of four major components:

- 1) Street cameras for observing traffic.
- 2) A local computing unit for traffic analysis.
- 3) A traffic signal controller for signal-phase decisions.
- 4) An optional cloud or central server for storage and long-term analysis.

The use of local processing is proposed to reduce dependence on continuous transmission of raw camera data to a remote server. The actual communication delay and computational performance must, however, be measured experimentally before making performance claims.

VI. SYSTEM ARCHITECTURE

The proposed architecture follows the sequence:

Camera Input → Local Processing → Decision Module → Signal Controller

Traffic information is first captured by roadside cameras. The local processing unit analyzes the available information and estimates the traffic state. If an emergency vehicle is detected with sufficient confidence, the decision module generates a priority request. The traffic signal controller then determines the appropriate signal transition.

A cloud or central server can optionally be used for storing aggregated information and performing long-term analysis. It is not required for every real-time control decision.

VII. SYSTEM FLOWCHART

The proposed processing sequence is shown in Fig. 1.

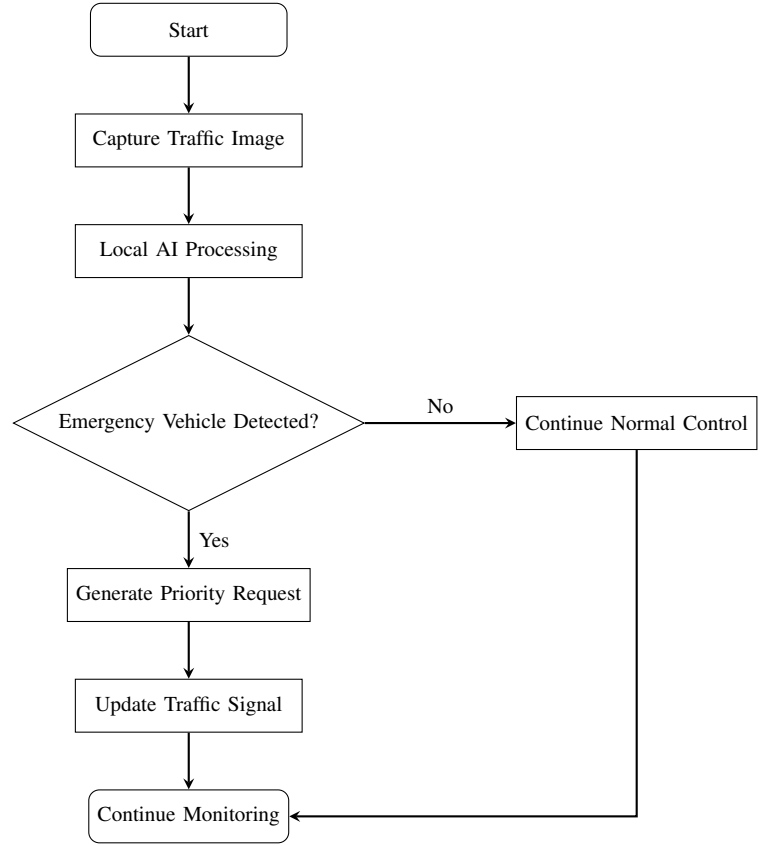


Fig. 1. Flowchart of the Proposed Traffic Signal Control System

VIII. MATHEMATICAL MODEL

The traffic state can be represented using measurable variables such as vehicle count and queue length. A simple vehicle-density measure can be expressed as

$$D = \frac{N_v}{L} \quad (1)$$

where N_v represents the number of detected vehicles and L represents the monitored road-section length.

An emergency priority decision can be represented conceptually as

$$P = \begin{cases} 1, & \text{if an emergency vehicle is detected with sufficient confidence} \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

When $P = 1$, the controller can evaluate an emergency-priority signal transition subject to safety and traffic-control constraints.

These equations represent a conceptual model. Actual thresholds and detection-confidence values must be established through testing.

IX. IMPLEMENTATION FRAMEWORK

A practical implementation can be divided into the following stages.

A. Traffic Image Acquisition

Roadside cameras continuously provide images or video frames of the intersection. The camera position should provide sufficient visibility of relevant lanes.

B. Local Processing

The local computing unit receives camera information and performs vehicle detection and traffic-state estimation. Machine-vision techniques can be considered for extracting vehicle-related information, as demonstrated in previous adaptive traffic-signal research [2].

C. Emergency Vehicle Detection

An emergency vehicle detection module can be designed to identify visual characteristics associated with emergency vehicles. A practical implementation should evaluate detection accuracy under different lighting, weather, vehicle-density, and camera-position conditions.

D. Signal Control

When an emergency vehicle is identified, the decision module sends a priority request to the traffic signal controller. The controller should perform a safe transition rather than abruptly changing conflicting signal states.

X. COMPARISON WITH EXISTING APPROACHES

Table I summarizes the main characteristics of the approaches discussed in the literature.

TABLE I
COMPARISON OF TRAFFIC SIGNAL CONTROL APPROACHES

Approach	Main Information Used	Main Focus
Deep RL based control [1]	Traffic-state information	Adaptive control
Machine vision based control [2]	Vehicle and queue information	Adaptive signal timing
Emergency priority algorithm [3]	Emergency vehicle requests	Emergency vehicle priority
Emergency vehicle-centered control [4]	Emergency vehicle and traffic conditions	Emergency vehicle delay and traffic impact
Proposed architecture	Camera-based traffic and emergency information	Local processing with emergency priority

The proposed architecture combines ideas from adaptive traffic observation and emergency vehicle priority. It should be treated as a design proposal until implemented and experimentally evaluated.

XI. EVALUATION PLAN

Since this paper presents a proposed architecture rather than a completed experimental system, numerical performance results are not claimed.

A future implementation can be evaluated using the following parameters:

- Emergency vehicle detection accuracy.
- Emergency vehicle waiting time.
- Average vehicle delay.
- Queue length.
- Signal decision latency.
- Detection performance under different traffic conditions.

The system can be evaluated first in a traffic simulator and later using a controlled prototype or field test, subject to appropriate safety and regulatory requirements.

XII. ADVANTAGES

The proposed architecture has the following potential advantages:

- Traffic information can be processed close to the intersection.
- The architecture can combine normal traffic adaptation with emergency vehicle priority.
- Only selected information needs to be forwarded to a central server when cloud storage is used.
- The architecture can be evaluated incrementally using simulation and prototype testing.

XIII. LIMITATIONS

The proposed architecture has several limitations.

Camera-based detection can be affected by poor lighting, weather, occlusion, camera position, and traffic density. Local processing also requires suitable computing hardware. Emergency vehicle detection errors could result in inappropriate priority requests, so safety constraints are necessary.

The proposed paper does not claim real-world deployment or measured performance.

XIV. FUTURE SCOPE

Future work can focus on implementing and testing the proposed architecture using a traffic simulator and a controlled prototype.

Further research can investigate improved emergency vehicle detection, multi-intersection coordination, vehicle-to-infrastructure communication, and adaptive signal-control algorithms. Additional testing can also examine how emergency priority affects normal traffic at neighboring intersections.

XV. CONCLUSION

This paper presented a conceptual architecture for smart traffic signal control using local AI-assisted traffic observation and emergency vehicle priority. Existing research demonstrates the use of machine vision and intelligent methods for adaptive traffic signal control, while other studies investigate emergency vehicle preemption.

The proposed architecture combines these concepts into a system in which roadside cameras provide information to a local processing unit, which can generate a priority request for the traffic signal controller. No experimental performance values are claimed because the proposed architecture has not been experimentally evaluated in this sample paper. Future implementation and controlled testing are required to determine detection accuracy, response time, and the effect on normal traffic.

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